Ashkan Moradi

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Education

Jan 2025 – Present	**Ph.D. student in Telecommunication, Institut national de la recherche scientifique (INRS),** Montreal, Canada. **Thesis title**: *Securing the Metaverse Through User Access Control.*
Sep 2018 – Feb 2022	**M.Sc. Artificial Intelligence and Robotics, Shahid Beheshti University,** Tehran, Iran. **Thesis title**: *Performance Improvement of Spoken Language Identification System via Fusion of the Acoustic and Phonetic Information based on learning methods*, (Grade: A+). **GPA**: 16.23 out of 20.
Sep 2013 – Jan 2018	**B.Sc. Computer Software Engineering, Sadjad University of Technology,** Mashhad, Iran. **Thesis title**: *Designing and Developing a Website about Machine Learning*, (Grade: A+). **GPA**: 16.18 out of 20.

Publications

1. Alipour Esgandani, S., Shekofteh, Y., & **Moradi, A.** (2024). Apedm: A new voice casting system using acoustic–phonetic encoder-decoder mapping. *Multimedia Tools and Applications*, 1–25.
2. **Moradi, A.**, & Shekofteh, Y. (2023). Spoken language identification using a genetic-based fusion approach to combine acoustic and universal phonetic results. *Computers and Electrical Engineering, 105*, 108549.
3. **Moradi, A.**, Shekofteh, Y., & Zarei, S. (2021). A genetic-based fusion approach of persian and universal phonetic results for spoken language identification. In *2021 11th international conference on computer engineering and knowledge (iccke)* (pp. 175–181). IEEE.

Work Experience

Teaching Assistant

2020	**Digital Speech Processing,** Shahid Beheshti University, Tehran, Iran.
	Open Banking, Tarbiat Modares University, Tehran, Iran.
2017	**Artificial Intelligence,** Sadjad University of Technology, Mashhad, Iran.
	Algorithms Design and Analysis, Sadjad University of Technology, Mashhad, Iran.
	Data Structures, Sadjad University of Technology, Mashhad, Iran.
2016 – 2017	**Operating Systems,** Sadjad University of Technology, Mashhad, Iran.
	Advanced Programming, Sadjad University of Technology, Mashhad, Iran.

Work Experience (continued)

2015 — **Introduction to Programming,** Sadjad University of Technology, Mashhad, Iran.

Professional Experience

Nov 2023 – Nov 2024 — **Tajan System**, .NET Developer, Tehran, Iran.

Jan 2022 – Sep 2023 — **Techofen**, R&D Manager, Tehran, Iran.

Mar 2020 – Oct 2023 — **Techofen**, Chief Technology Officer (CTO), Tehran, Iran.

May 2019 – Mar 2020 — **Saman Insurance Co.**, Full Stack Developer, Tehran, Iran.

Nov 2018 – May 2019 — **Saramad Holding**, Full Stack Developer, Tehran, Iran.

Projects and Research

2026 — **Speaker Attribute Prediction**: Benchmarking Foundation Models for Cross-Domain Speaker Profiling

2025 — **Speech Deepfake Detection**: Towards Lightweight On-Device Audio Deepfake Detection using Squeezeformers

2024 — **Employee Performance Evaluation System**: In this project, we developed an enterprise system to evaluate employees across various departments within a company. It featured monthly, seasonal, and yearly report cards for performance comparison, along with tools like polling, peer feedback, goal tracking, and customizable evaluation templates.

2022 — **Engine Fault Detection Using Engine Sound**: In this project, we are examining and using different acoustic features to detect engine faults in an unsupervised manner.

2020 – 2022 — **Performance Improvement of Spoken Language Identification System via Fusion of the Acoustic and Phonetic Information based on learning methods**: This was my M.Sc. thesis project, for which I developed an acoustic-based x-vector system and a phonetic system based on the Allosaurus universal phone recognizer. Finally, I proposed two different fusion methods based on a genetic algorithm to fuse the mentioned system.

2020 — **Sentiment Analysis on SOHA system of Shahid Beheshti University**: The goal of this project was to classify comments of SOHA systems into three groups: positive, negative, and neutral. Since these comments were not labeled, I used a lexicon-based approach to classify them.

Implementation of "Many Objective Cooperative Bat Searching Algorithm": In this project, I implemented the "Many objective cooperative bat searching algorithm" paper to optimize problems with different objective functions.

2019 — **Credit Card Fraud Detection Using Hmm**: This project aimed to detect credit card fraudulent transactions using the Hidden Markov Model and different clustering methods.

Multiclass Semantic Segmentation: The goal of this project was to implement multi-class semantic segmentation using Fully Convolutional Networks (FCN) on the CamVid dataset.

2018 — **Constrained Clustering**: The goal of this project was to add some constraints to the K-means clustering algorithm. The project was done based on active-learning methods.

Projects and Research (continued)

2017 — **Speaker Gender Detection**: The project was based on some speech-related features such as MFCC, pitch period, short-time energy, and spectral centroid.

Skills

Languages — **Persian**: Native proficiency, **English**: Full professional proficiency, **French**: Elementary proficiency, **Spanish**: Elementary proficiency.

Coding — Python, PyTorch, TensorFlow, Matlab, C#, C++, JS, .Net, LaTeX.

AI — Speaker Profiling, Language and Speaker Recognition, Speech and Signal Processing, Speech Foundation Models, Machine/Deep Learning, Natural Language Processing.

Other — Kaldi, Linux, Docker, Team Management, Project Management, Time Management, System Design, Full Stack Developer, Rest API.

Certifications and Training

2023
- **Custom and Distributed Training with TensorFlow**, by Coursera.
- **Custom Models, Layers, and Loss Functions with TensorFlow**, by Coursera.

2022
- **DeepLearning.AI TensorFlow Developer Specialization**, by Coursera.
- **Sequences, Time Series and Prediction**, by Coursera.
- **Natural Language Processing in TensorFlow**, by Coursera.
- **Machine Learning Specialization**, by Coursera.
- **Machine Learning: Clustering & Retrieval**, by Coursera.
- **Convolutional Neural Networks in TensorFlow**, by Coursera.
- **Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning**, by Coursera.
- **Python for Deep Learning: Build Neural Networks in Python**, by Udemy.
- **Machine Learning: Classification**, by Coursera.
- **Machine Learning From Basic to Advanced**, by Udemy.
- **NLP Course for Beginner**, by Udemy.

2018
- **Machine Learning: Regression**, by Coursera.
- **Applied Machine Learning in Python**, by Coursera.
- **Machine Learning Foundations: A Case Study Approach**, by Coursera.
- **Applied Plotting, Charting & Data Representation in Python**, by Coursera.
- **Introduction to Data Science in Python**, by Coursera.

Miscellaneous

2019 — **Best data mining project of the year award winner** at the university.

2016 — **ACM/ICPC 2016 Technical Committee**, by Sadjad University of Technology, Mashhad, Iran.

2015 — **ACM/ICPC 2015 Excecutive Committee**, by Sadjad University of Technology, Mashhad, Iran.